Summary in Graph

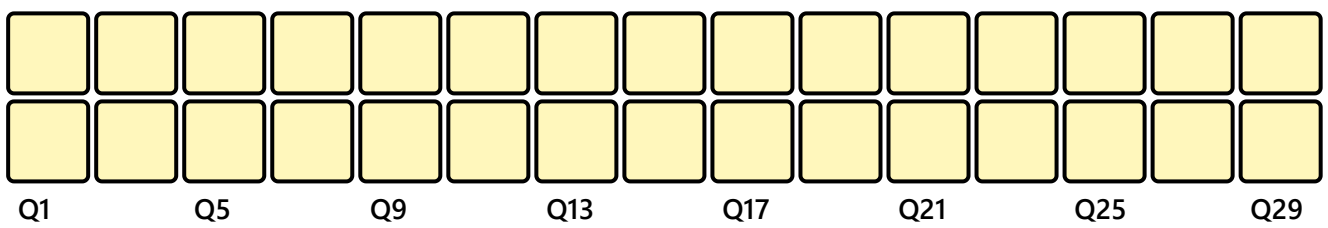
Exam Summary (GO Classes CS Test Series 2026 | Operating Systems | Subject Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (20 + 10)
Total Marks:	40 (20 + 20)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (56) out of 560 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A system is composed of four processes, $\{P1, P2, P3, P4\}$, and three types of consumable resources, $\{R1, R2, R3\}$. There is 1 unit of each type of resource.

- P1 acquired 1 unit of R1 and requested 1 unit of R3.
- P2 acquired 1 unit of R3 and requested 1 unit of R1.
- P3 requested 1 unit of R2.

- P4 acquired 1 unit of R2 and 1 requested 1 unit of R3.

Which of the following processes are in deadlock?

A. $P1$
B. $P2$
C. $P3$
D. $P4$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 15sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a disk with the following characteristics:

- Number of surfaces: 8
- Number of tracks / surface: 512 K
- Number of bytes / track: 8 MB
- Number of sectors / track: 8 K
- On-disk cache: 16 MB

What is the size of the disk(in TeraBytes, $1\text{TB} = 2^{40}B$)?

Your Answer: | Correct Answer: 32 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A processor has a 32-bit virtual address space and 32 bit physical address space, and uses a 2-level page table. To load a process, the entire page table is first loaded into the main-memory. The page fault rate is 20%. The main-memory access time is 10 ns , while page fault service time is 1000 ns. What will be the effective access time for the main-memory?

A. 208 ns
B. 224 ns
C. 216 ns
D. 220 ns

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a computer system with a main memory of size 1 MB. A customized operating system of size 150 KB has been loaded in the lower side of the memory. The contiguous memory allocation (variable partition) scheme is used with the first-fit strategy. The processes arrive and terminate in the following sequence: P1 (250 KB) arrives, P2 (100 KB) arrives, P3 (150 KB) arrives, P2 terminates, P4 (220 KB) arrives. All the processes are loaded from the lowest address of the allocated hole. What is the smallest process size in KiloByte that can be denied for the memory allocation?

A. 100 KB
B. 155 KB
C. 154 KB
D. 254 KB

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Consider a demand-paging system where the degree of multiprogramming is currently fixed to four. The system utilizes 15% of the CPU and 5% of the disk in OS related activities. The system recently measured and determined the CPU and paging disk utilization for the processes. In which of the following cases the system can increase the degree of multiprogramming?

A. CPU utilization 13%; disk utilization 97%
B. CPU utilization 85%; disk utilization 3%
C. CPU utilization 13%; disk utilization 3%
D. CPU utilization 90%; disk utilization 10%

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a demand paging system with 3 page frames, initially all page frames are empty. LRU page replacement policy is used for the following page references.

$$1 \quad 2 \quad 3 \quad 4 \quad 1 \quad 6 \quad 3 \quad 6 \quad 1 \quad 8 \quad 4 \quad 1 \quad 8$$

The page fault rate, defined as the ratio of number of page faults to the number of memory references is(up to 2 decimal places)?.

Your Answer: | Correct Answer: 0.69 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A computer system uses a $32-$bit virtual address space with a page size of 4 KB. The physical memory is 256 MB in size. The system uses a single level page table, and each page table entry (PTE) is 20-bits long.
The virtual address $\mathrm{VA} = 0\mathrm{x}1234\mathrm{ABCD}$ maps to the page table entry with frame $0\mathrm{x}2\mathrm{F}3\mathrm{B}0$. Which of the following is the corresponding physical address (in hexadecimal)?

A. $0\mathrm{x}2\mathrm{F}3\mathrm{ABCD}$
B. $0\mathrm{x}2\mathrm{F}34\mathrm{BCD}$
C. $0\mathrm{x}2\mathrm{F}3\mathrm{B}0\mathrm{CD}$
D. $0\mathrm{x}2\mathrm{F}3\mathrm{BBCD}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A system uses a 12-bit virtual address space and a one-level page table. The page size is 16 bytes and the portion of the page table is shown below:

Page Number	Frame Number
67	100
150	500
255	900

What is the physical address corresponding to the virtual address 4095 (in decimal)?

Your Answer: | **Correct Answer: 14415** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A virtual memory system uses paging with segmentation. Process virtual addresses consist of a total of 32 bits and the page size is 1024 bytes. A page table may occupy at most one frame, and the size of a page table entry is 4 bytes. A single process may have up to 64 (2^6) segments. What is the maximum number of bits required to address a virtual memory segment?

Your Answer: | **Correct Answer: 26** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a set of five process, with the length of CPU burst time given in milliseconds:

Process	Burst Time	Arrival Time
P_1	5	4
P_2	3	1
P_3	1	2
P_4	7	2
P_5	4	3

The minimum achievable average waiting time for these four processes in a shortest remaining first scheduling (rounded to nearest integer) is _____ ms?

Your Answer: | **Correct Answer: 4** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements is/are True about semaphores?

A. Semaphores can be implemented in such a way that prevents busy waiting.
B. If there is at least one thread sleeping on a given semaphore, then the result of performing a Semaphore V() will be to wake up one of the sleeping threads.
C. Semaphores cannot be used for locking.
D. The pure semaphore interface does not allow querying for the current value of the semaphore.

Your Answer: | **Correct Answer: A;B;D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is associated with the User-level threading?

A. Concurrency
B. Shared Heap
C. Parallelism
D. Transparent to kernel

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Assume that the following processes are the only processes in a computer system. Given the following arrival time and CPU burst time (also called service time) for each process, what is the average waiting time if preemptive shortest job first scheduling is used?

Process	Arrival Time	CPU Burst Time
P_1	0	20
P_2	3	7
P_3	5	2
P_4	7	3

Your Answer: | Correct Answer: 4.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

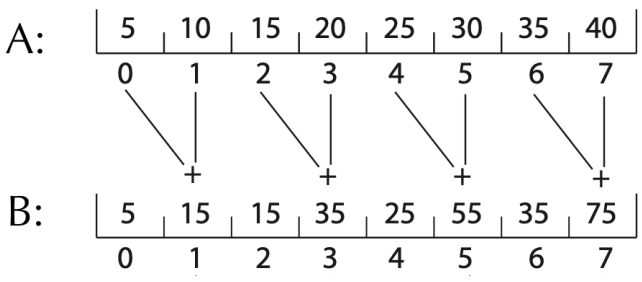
```
Void mystery()
{
    wait(S);
    wait(S);
    //critical section
    signal(S);
    signal(S);
}
```

Consider a mystery() function implementing the critical section where S is a counting semaphore initialized with some unknown value X. If there are 7 processes referring to the mystery function then the minimum value of X such that the system is in safe state?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider two arrays A and B shown below:



Process 1:	Process 2:
For i = 0 to n: B[i+1] = A[i] + A[i+1] i = i + 2;	For j = 0 to n: B[j] = A[j] j = j + 1;

Two concurrent processes P1 and P2 are working on the same shared array A and B . What is the minimum number of binary semaphores required to ensure the correct result in array B as shown?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following scenarios always causes a transition from user-mode to kernel-mode?

A. A packet is received from the network.
B. The application uses malloc() to allocate memory from the heap.
C. The timer goes off.
D. The application runs in critical section.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Consider the following pseudocode, executed concurrently by two processes to access a shared critical section:

```
while (flag == true)
    do nothing;
flag = true;
Execute Critical Section;
flag = false;
```

Which of the following statements are TRUE, assuming the flag is initially set to False?

A. The solution does not ensure mutual exclusion.
B. The solution may cause a deadlock.
C. The solution may cause starvation.
D. The solution satisfies the progress requirement.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements is/are True about synchronization?

A. Synchronization is used to prevent race conditions.
B. At least one process should be in the critical section at any time to satisfy progress.
C. Bounded waiting ensures the number of times processes are allowed to enter their critical sections after a process has made a request to enter its critical section.
D. Progress condition ensures that the decision of which process enters the critical section next cannot be postponed indefinitely.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider the following three threads in a concurrent program that uses semaphores Sem1, Sem2, and Sem3.

Thread 1	Thread 2	Thread 3

L1: sema_down(Sem3);
print("2");
sema_up(Sem2);
goto L1;
	L2: sema_down(Sem1);
	print("6");
	sema_up(Sem3);
	goto L2;
		L3: sema_down(Sem2);
		print("1");
		sema_up(Sem1);
		goto L3;

What are the initial values that can be given to the semaphores so that the threads cooperate to print a string that begins with 16216216216216 ?

A. $\text{sem}\,1 = 1, \text{sem}\,2 = 0, \text{sem}\,3 = 0$
B. $\text{sem}\,1 = 1, \text{sem}\,2 = 1, \text{sem}\,3 = 1$
C. $\text{sem}\,1 = 0, \text{sem}\,2 = 1, \text{sem}\,3 = 0$
D. $\text{sem}\,1 = 1, \text{sem}\,2 = 1, \text{sem}\,3 = 0$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A resource pool allows exactly one process to pass at a time. There are m processes currently inside the resource pool who must exit before any of the n waiting processes outside can enter. To implement this two counting semaphores allow_entry(initialized to 0), allow_exit (initialized to m) and additionally one binary semaphore mutex(initialized to 1) is used. processes use functions get_in() and get_off() to enter and exit the resource pool respectively.

void get_off(){	void get_in(){

1. wait(allow_exit);
2. wait(*mutex*);
3. // critical section
4. If allow_exit == 0:
allow_entry = n;
5. signal(*mutex*);
}
	1. wait(allow_entry);
	2. wait(*mutex*);
	3. // critical section
	4. If allow_entry == 0:
	allow_exit = m;
	5. signal(*mutex*);
	}

Which of the following are correct about this solution?

A. Mutual exclusion is satisfied.

B. All the processes first enter and then all exit from the resource pool.

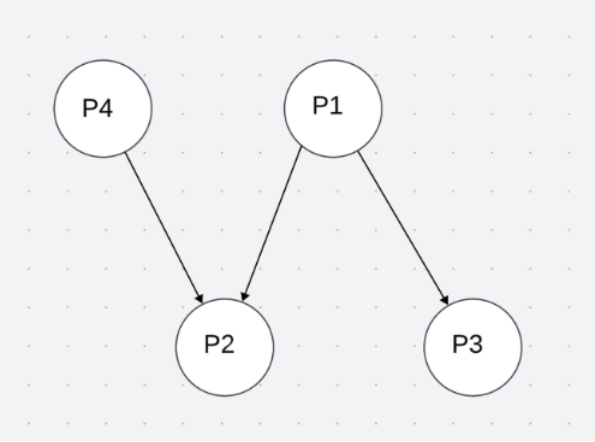
C. Exchanging line 1 and 2 in both functions can cause deadlock.

D. 'If' condition can be used after 'signal' in both functions.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A system is concurrently running four threads $\mathrm{P1}, \mathrm{P2}, \mathrm{P3}$ and $\mathrm{P4}$, of a process P which share a critical section. The order of execution of their critical section is given by the topological ordering as shown below:



The minimum number of binary semaphores required to enforce this execution sequence?

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider four processes A, B, C, and D scheduled on a CPU as per round robin algorithm with a time quantum of 2 units. The processes arrive in the order A, B, C, D, all at time $t = 0$. There are exactly two context-switches from A to B, exactly 1 context switch from C to D and exactly two context switches from D to A. There is no context switch between B and D. Which one of the following is NOT possible as the burst time of these processes?

A. $\mathrm{A} = 8; \mathrm{B} = 4; \mathrm{C} = 4; \mathrm{D} = 6$
B. $\mathrm{A} = 8; \mathrm{B} = 3; \mathrm{C} = 3; \mathrm{D} = 6$
C. $\mathrm{A} = 8; \mathrm{B} = 5; \mathrm{C} = 4; \mathrm{D} = 6$
D. $\mathrm{A} = 7; \mathrm{B} = 3; \mathrm{C} = 3; \mathrm{D} = 5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the system that supports a virtual address of 58 bits and physical address of 32 bits, the page table is organised in 3 levels. Each page table entry is 4 bytes, consisting of 4 fields, frame number, one dirty bit, one valid bit and some protection bits. The length of the frame field in the page table entry is _____ bits?

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A processor uses a 32-bit logical address and a 32-bit physical address space. The page size is 4 KiloBytes, and the size of each page table entry (PTE) is 1 byte. The virtual address is divided as follows:

$6-\text{bit}$	$6-\text{bit}$	$8-\text{bit}$	$12-\text{bit}$
$1-\text{level}$	$2-\text{level}$	$3-\text{level}$	$\text{page}-\text{offset}$

The number of bits required for addressing the next level page table(or page frame) in the page table entry of the first, second and third level page tables respectively are?

A. $26, 26, 20$

B. $12, 20, 20$
C. $12, 12, 12$
D. $26, 24, 20$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A processor supports a 32-bit logical address space and a 32-bit physical address space. The page size is 1 KB (page offset $= 10$ bits), and each page-table entry (PTE) is 8 bytes. If a process uses only one-fourth(contiguous) of its total logical address space, then how much memory can it save(in MB, up to 2 decimal places) in page tables with multilevel paging?

L1	L2	L3	offset

$32 - \text{bit}$

Your Answer: | Correct Answer: 23.94 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Suppose that you have a UNIX file system where the disk block size is 1 KB , and an inode takes 128 bytes. Disk addresses take 32 bits, and the inode contains space for 64 bytes of data, 8 direct addresses, one indirect, one double-indirect and one triple-indirect(the rest of the space in the inode is taken up with other information such as ownership and protection). How much space (including overhead) a 1025 bytes file takes in the disk?(in bytes)

Your Answer: | Correct Answer: 1152 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A Computer system supports two different file systems. In both the systems disk block size is 512 Bytes and inode is 1 KB in size. Consider a file of 50 blocks in size stored on disk, a new block is inserted in the middle(between the 25th and 26th block) of the file. Assume enough space is available on the disk, currently nothing is buffered in the main-memory and after insertion the result must be stored back on the disk. if P and Q are the number of disk accesses required to perform the insert operation in a contiguous and linked allocation file system respectively then $\mathrm{P} + \mathrm{Q} =$ _____ ?

A. 53
B. 78
C. 82
D. 86

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A Unix-like operating system executes the following C code fragment in a process.

```
int main(){
    fork();
    fork();
    fork();
5.  fork();
}
```

Assume in this system the number of successful fork system calls are characterised by the uniform random variable X, Find $P(X > 5)$?

Your Answer: | Correct Answer: 0.625 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Which of the following are is/True about the Bankers algorithm?

A. Banker's algorithm guarantees a safe state given a safe sequence.
B. There always exists a safe sequence for any system state.
C. The algorithm can approve a resource request regardless of the given safe sequence.
D. If a resource request is not approved, and the resource request is processed, then the system necessarily will enter in deadlock.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a system with N process and $\mathrm{K}, (> \mathrm{N})$ resources of the same type. What could be the maximum requirement of each process such that the system is in safe state?

A. $\leq K - 1$
B. $\leq \left\lfloor \frac{N}{K} \right\rfloor + 1$
C. $\leq \left\lceil \frac{K}{N} \right\rceil$
D. $\leq \left\lfloor \frac{K-1}{N} \right\rfloor + 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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